Substring in Java

02 January 2026 12:17

A part of a string is called substring. In other words, substring is a subset of another string. Java String Glass provides the built in substring() method that extract a substring from a given string by using the index values passed as an argument. In case of substring() method start index is inclusive and end index is exclusive.

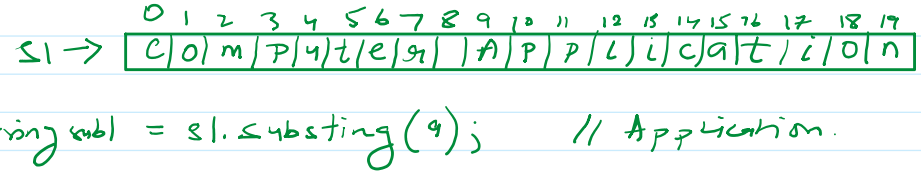
You can get substrings from the given string object by one of the two methods:

1. **public String substring(int startIndex):** This method returns a new String object containing the substring of the given string from the specified start index (inclusive). The method throws an **StringIndexOutOfBoundsException** when the start index is larger than the length of a string or less than 0.

2. **public String substring(int startIndex, int endIndex):** This method returns new String object containing the substring of the given string from specified start index to end index. The method throws an **StringIndexOutOfBoundsException** when the start index is less than 0 or start index is greater than end index or end index is greater than the length of the String.

   In case of String:
   - startIndex: inclusive
   - endIndex: exclusive

```
String s1 = "Computer Application";
```



String sub1 = s1.substring(9); // Application.

```
String s1 = "Computer Application";
String sub1 = s1.substring(9);
sub1
"Application"   (String)
String sub2 = s1.substring(-1);
Exception: java.lang.StringIndexOutOfBoundsException (Range [-1, 20) out of bounds for
length 20)

String sub2 = s1.substring(20);
sub2
""   (String)
```

← empty string because our string length is 20.

```
sub2 = s1.substring(21);
Exception: java.lang.StringIndexOutOfBoundsException (Range [21, 20) out of bounds for
length 20)


s1
"Computer Application"   (String)
```

```
sub2 = s1.substring(3, 8);
sub2
"puter"   (String)

sub2 = s1.substring(-3, 8);
Exception: java.lang.StringIndexOutOfBoundsException (Range [-3, 8) out of bounds for
length 20)

sub2 = s1.substring(3, 20);
sub2
"puter Application"   (String)
sub2 = s1.substring(3, 21);
Exception: java.lang.StringIndexOutOfBoundsException (Range [3, 21) out of bounds for
length 20)
```

-ve indexing is not allowed in Java

specified index is greater the s1.length()